

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

STC 111 — Past Questions & Materials

by Renance

Course: STC 111

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 2

Levels: ND1

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Paper 1 of 2 — STC 111 NEW

Level ND1 • Second Semester • Handout

STC 111

EXPERIMENT 2

TITLE: IDENTIFICATION OF METALLIC IONS USING FLAME TEST TECHNIQUE

AIM:

DEFINITION

A flame test is a qualitative test applied to determine the identity of metals and their ions in compounds. This can be done by observing the colors of metal ions based on the emission spectrum. On heating, every metal ion has its unique color at a specific wavelength.

THEORY

When metals are heated on flame, their electrons jump to higher energy levels. On returning to their original energy level, they emit energy that can be seen in the form of colors by the naked eye. Based on their colors, metal ions can be easily identified in a compound. Some of them do not show colors because they are colorless. When performing the flame test to determine the presence of metal ions in the sample, there are substances responsible for the contamination of flame tests.

These are the sources of contaminants:

- Sodium metal ions (mix the yellow color with other metals)
- Sweat from hands (contain several metals)
- Dust impurities
- Organic materials
- Unwashed nichrome wire

The major contaminant is sodium metal. It gives a yellow color flame that masks the colors of other metals. Cobalt blue glass is used to minimize the yellow color of sodium. In this way, minimization of the yellow color of the flame allows us to see other colors of the metal.

APPARATUS:

Nichrome/platinum wire, Bunsen burner, test tubes, test tube rack,

REAGENTS:

Dil. HCl, salt samples to be tested, distilled water

PROCEDURE

- Clean a platinum or nichrome (a nickel-chromium alloy) wire by dipping it into concentrated hydrochloric acid and then holding it in a hot (non-luminous) Bunsen flame.

- Repeat this until the wire doesn't produce any colour in the flame.
- When the wire is clean, moisten it again with some of the acid.
- Make a small amount of concentrated solution of the samples to be tested inside different test tubes and carefully label them.
- Dip the clean wire into the test tube containing the sample you are testing so that some sticks to the wire.
- Place the wire back in the flame again, carefully notice and record the colour change.

DIAGRAM

NB.

If the flame colour is weak, it is often worthwhile to dip the wire back in the acid again and put it back into the flame as if you were cleaning it. You often get a very short but intense flash of colour by doing that.

RESULT

IONS COLOUR

Li+

Na+

K+

Ca 2+

Sr 2+

Ba 2+

Cu 2+

OBSERVATION:

PRECAUTIONS

SOURCES OF ERROR

RECOMMENDATION

CONCLUSION

EXPERIMENT 3

TITLE: INVESTIGATION OF THE REACTIVITY OF GROUP 2 METALS

AIM:

DEFINITION:

THEORY:

APPARATUS:

* Bunsen burner, thermometer, Electronic balance, Measuring cylinders, Pipettes, Safety goggles, gloves, and lab coat

REAGENTS:

* Group 2 metals: Magnesium (Mg), Calcium (Ca) ,

* Hydrochloric acid (HCl), Distilled water.

PROCEDURE

REACTION WITH WATER

* Cut the Group 2 metals into small pieces, approximately 1-2 cm in length.

* Add a small piece of each metal to a separate container of distilled water.

* Observe the reaction and record any observations, such as gas evolution, color change, or temperature increase.

REACTION WITH ACID

* Prepare a solution of acid (1M HCl) by measuring out the required volume using a measuring cylinder.

* Add a few drops of acid to a small piece of each metal using a pipette.

* Observe the reaction and record any observations, such as gas evolution, color change, or temperature increase.

REACTION WITH HEAT

* Heat a small piece of each metal using a Bunsen burner or other heat source.

* Observe the reaction and record any observations, such as melting, vaporization, or color change.

DIAGRAM :

OBSERVATION:

RESULT:

PRECAUTIONS:

1. Wear safety goggles, gloves, and lab coat to protect against acid splashes and heat.
2. Handle acids with care, as they can cause burns and other injuries.
3. Use the heat source carefully, as it can cause burns and other injuries.

SOURCES OF ERROR

RECOMMENDATION:

CONCLUSION:

EXERCISE

1. Compare the reactivity of the Group 2 metals based on observations and recorded data.
2. Determine the reactivity trend of the Group 2 metals based on their reactions with water, acid, and heat.

EXPERIMENT 4

TITLE: INVESTIGATION OF THE REACTIVITY OF TRANSITION METALS AND THEIR

COMPOUND

AIM:

DEFINITION:

THEORY:

APPARATUS

Eye protection, Test tubes, bar magnet, Dropping pipette

REAGENTS

Samples of some transition metals, such as copper, iron and zinc.

Copper(II) sulfate 0.01 mol dm⁻³, Iron(III) chloride 0.1 mol dm⁻³, Ammonia solution 2 mol dm⁻³

PROCEDURE

- * Test the metal samples for hardness and ability to bend without breaking. Record your observations qualitatively.
- * Place the metal samples close to a bar magnet to find out which samples are magnetic. Record your observation
- * Set up an experiment to see if the metals react with water. (This may need to be left for some time).
- * Take a small sample of a solution of copper(II) sulfate (approximately 2cm³), add ammonia solution to it a few drops at a time. Record your observations.
- * Add ammonia solution until there is no further change.
- * Repeat with the other solutions of transition metal compounds.

DIAGRAM :

OBSERVATION:

RESULT:

PRECAUTIONS:

SOURCES OF ERROR

RECOMMENDATION:

CONCLUSION:

EXERCISE

1. Describe the physical properties of transition metals.
2. How do transition metals react with water?
3. What properties do the compounds of transition metals have in common?

EXPERIMENT 5

TITLE: SEPARATION OF MIXTURE OF SALT AND SAND

AIM:

DEFINITION/THEORY:

EQUIPMENTS/APPARATUS

Eye protection

Beaker, 250 cm³

Glass stirring rod

Filter funnel

Filter paper

Conical flask, 250 cm³

Evaporating basin

Bunsen burner

Heat resistant mat

Tripod

Gauze

CHEMICAL/MATERIAL :

NaCl, Distilled water, sand

PROCEDURE:

1. Weigh the empty Erlenmeyer flask. Record data.
2. Using the spatula, take one spoonful of salt. Use weigh paper to weigh the salt. Record data.
3. Next, take 3 spoonfuls of sand, use weigh paper to weigh sand. Record data.
4. Mix the salt and the sand. Pour into the 150 mL beaker. (It should cover the bottom of the beaker.)
5. Add approximately 10-20 mL warm/hot water to the salt/sand mixture and stir with the glass stirring rod until the salt is fully dissolved. (Warm or hot water will help in dissolving the salt quickly.)
6. Prepare a filter cone with the filter paper.
7. Place filter paper cone in the funnel. Next, place the funnel in the top of the Erlenmeyer flask and pour the mixture into the Erlenmeyer flask. (Filtering process may take a while.)
8. While the saltwater mixture is filtering into the flask, place hot plate on a flat surface.
9. After the salt and sand is filtered, place Erlenmeyer flask in the center of the hot plate. Turn the knob on to the #4 setting. Caution: HOT SURFACE! Watch carefully as the water begins to evaporate and leaves the isolated salt.
10. When all of the water is evaporated, let it cool completely and then place flask on scale to weigh. Record data.

11. Calculate the percent of salt from the mixture. Record data.

12. Rinse all equipment used and discard the sand and salt.

DIAGRAM

RESULT

S/N DATA Mass in grams

CALCULATION:

OBSERVATION:

PRECAUTIONS

SOURCES OF ERROR

RECOMMENDATION

CONCLUSION

EXPERIMENT 6

TITLE: PREPARATION OF IRON SULPHIDE

AIM:

DEFINITION:

INTRODUCTION:

This demonstration or class experiment shows the exothermic reaction of two elements, iron and sulfur, to form the compound, iron sulfide. The two solids are mixed and heated in a test tube (or ignition tube). The reaction can be used to illustrate elements, mixtures and compounds.

EQUIPMENT/APPARATUS

Eye protection

Balance (1 or 2 decimal places)

Test tube made from borosilicate glass (eg Pyrex)

Bunsen burner

Heat resistant mat/fume cupboard

Clamp stand and clamp

Spatulas x2

Small bar magnet

Watch glass

Filter paper (2 pieces or 2 weighing boats)

Mineral wool (or mineral fibre)

Test tube tongs

CHEMICALS/REAGENTS

Iron powder (potential IRRITANT)

Sulfur – finely powdered roll or flowers

PROCEDURE

1. Prepare a mixture containing iron powder and sulfur powder in the ratio 7:4 by mass. Do this by weighing out 7 g of iron powder and 4 g of finely powdered sulfur onto separate pieces of filter paper (or use weighing boats).

2. Mix the two powders by pouring repeatedly from one piece of paper to the other until a homogeneous mixture (by appearance) is obtained.

Note the appearance of the pure elements and the mixture.

3. Demonstrate that iron can be separated from the mixture by physical means. Do this by wrapping the end of a small bar magnet in a paper tissue, and dipping it into a teaspoon sized heap of the mixture on a watch glass. The iron will be attracted, but the sulfur remains on the watch glass.

4. Place about 2 g of the mixture into a borosilicate test tube.

Insert a plug of mineral wool (mineral fibre) into the mouth of the test tube. Clamp the test tube.

5. Heat the powder mixture at the base of the test tube – gently at first and then more strongly (use a blue flame throughout). Heat until an orange glow is seen inside the test tube. Immediately stop heating.

6. Allow the test tube to cool down.

7. Once cool, break open the test tube to show the appearance of the product, iron(II) sulfide. The test tube can be broken open using a pestle and mortar. It is advisable to wear protective gloves.

DIAGRAM:

OBSERVATION:

RESULT:

SAFETY NOTE:

* Wear eye protection throughout and ensure that the lab is well ventilated.

* Iron powder, Fe(s), (potential IRRITANT) – this can cause severe irritation in eyes because the iron oxidises rapidly in the saline environment

* Iron powder is preferred to iron filings. If fine sulfur powder is mixed with iron filings, it is difficult to obtain a consistent mix, because the two solids can separate.

* Sulfur, S(s) Roll sulfur or flowers of sulfur should be finely powdered using a pestle and mortar.

* Sulfur dioxide, SO₂(g), (TOXIC) is formed if the sulfur catches fire. So, to prevent this, use cotton wool to plug the test tube.

* This reaction can be carried out as a demonstration or class experiment in a well ventilated laboratory provided that the instructions provided here are strictly adhered to.

*The reaction can be carried out in borosilicate glass test tubes as a demonstration or in smaller (ignition) tubes by students.

*The reaction must not be carried out on tin lids in the open laboratory as is suggested in some sources. The sulfur may boil or burn releasing sulfur dioxide which is a TOXIC and CORROSIVE gas and may trigger an asthmatic attack.

RECOMMENDATION

CONCLUSION:

EXERCISE/ACTIVITY

Using a bar magnet, check and comment on the magnetic property of the product form.

Paper 2 of 2 — STC 111 MANUAL

Level ND1 • Second Semester • Handout

EXPERIMENT 5

TITLE: INVESTIGATION OF THE REACTIVITY OF GROUP 2 METALS

AIM:

DEFINITION:

THEORY:

APPARATUS:

* Bunsen burner, thermometer, Electronic balance, Measuring cylinders, Pipettes, Safety goggles, gloves, and lab coat

REAGENTS:

* Group 2 metals: Magnesium (Mg), Calcium (Ca) ,

* Hydrochloric acid (HCl), Distilled water.

PROCEDURE

REACTION WITH WATER

* Cut the Group 2 metals into small pieces, approximately 1-2 cm in length.

* Add a small piece of each metal to a separate container of distilled water.

* Observe the reaction and record any observations, such as gas evolution, color change, or temperature increase.

REACTION WITH ACID

* Prepare a solution of acid (1M HCl) by measuring out the required volume using a measuring cylinder.

* Add a few drops of acid to a small piece of each metal using a pipette.

* Observe the reaction and record any observations, such as gas evolution, color change, or temperature increase.

REACTION WITH HEAT

* Heat a small piece of each metal using a Bunsen burner or other heat source.

* Observe the reaction and record any observations, such as melting, vaporization, or color change.

DIAGRAM :

OBSERVATION:

RESULT:

PRECAUTIONS:

1. Wear safety goggles, gloves, and lab coat to protect against acid splashes and heat.

2. Handle acids with care, as they can cause burns and other injuries.
3. Use the heat source carefully, as it can cause burns and other injuries.

SOURCES OF ERROR

RECOMMENDATION:

CONCLUSION:

EXERCISE

1. Compare the reactivity of the Group 2 metals based on observations and recorded data.
2. Determine the reactivity trend of the Group 2 metals based on their reactions with water, acid, and heat.

EXPERIMENT 6

TITLE: INVESTIGATION OF THE REACTIVITY OF TRANSITION METALS AND THEIR COMPOUND

AIM:

DEFINITION:

THEORY:

APPARATUS

Eye protection, Test tubes, bar magnet, Dropping pipette

REAGENTS

Samples of some transition metals, such as copper, iron and zinc.

Copper(II) sulfate 0.01 mol dm⁻³, Iron(III) chloride 0.1 mol dm⁻³, Ammonia solution 2 mol dm⁻³

PROCEDURE

- * Test the metal samples for hardness and ability to bend without breaking. Record your observations qualitatively.
- * Place the metal samples close to a bar magnet to find out which samples are magnetic. Record your observation
- * Set up an experiment to see if the metals react with water. (This may need to be left for some time).
- * Take a small sample of a solution of copper(II) sulfate (approximately 2cm³), add ammonia solution to it a few drops at a time. Record your observations.
- * Add ammonia solution until there is no further change.
- * Repeat with the other solutions of transition metal compounds.

DIAGRAM :

OBSERVATION:

RESULT:

PRECAUTIONS:

SOURCES OF ERROR

RECOMMENDATION:

CONCLUSION:

EXERCISE

1. Describe the physical properties of transition metals.
2. How do transition metals react with water?
3. What properties do the compounds of transition metals have in common?

EXPERIMENT 7

TITLE: PREPARATION OF IRON SULPHIDE

AIM:

DEFINITION:

INTRODUCTION:

This demonstration or class experiment shows the exothermic reaction of two elements, iron and sulfur, to form the compound, iron sulfide. The two solids are mixed and heated in a test tube (or ignition tube). The reaction can be used to illustrate elements, mixtures and compounds.

EQUIPMENT/APPARATUS

Eye protection

Balance (1 or 2 decimal places)

Test tube made from borosilicate glass (eg Pyrex)

Bunsen burner

Heat resistant mat/fume cupboard

Clamp stand and clamp

Spatulas x2

Small bar magnet

Watch glass

Filter paper (2 pieces or 2 weighing boats)

Mineral wool (or mineral fibre)

Test tube tongs

CHEMICALS/REAGENTS

Iron powder (potential IRRITANT)

Sulfur – finely powdered roll or flowers

PROCEDURE

1. Prepare a mixture containing iron powder and sulfur powder in the ratio 7:4 by mass. Do this by weighing out 7 g of iron powder and 4 g of finely powdered sulfur onto separate pieces of filter paper (or use weighing boats).
2. Mix the two powders by pouring repeatedly from one piece of paper to the other

until a homogeneous mixture (by appearance) is obtained.

Note the appearance of the pure elements and the mixture.

3. Demonstrate that iron can be separated from the mixture by physical means. Do this by wrapping the end of a small bar magnet in a paper tissue, and dipping it into a teaspoon sized heap of the mixture on a watch glass. The iron will be attracted, but the sulfur remains on the watch glass.

4. Place about 2 g of the mixture into a borosilicate test tube.

Insert a plug of mineral wool (mineral fibre) into the mouth of the test tube. Clamp the test tube.

5. Heat the powder mixture at the base of the test tube – gently at first and then more strongly (use a blue flame throughout). Heat until an orange glow is seen inside the test tube. Immediately stop heating.

6. Allow the test tube to cool down.

7. Once cool, break open the test tube to show the appearance of the product, iron(II) sulfide. The test tube can be broken open using a pestle and mortar. It is advisable to wear protective gloves.

DIAGRAM:

OBSERVATION:

RESULT:

SAFETY NOTE:

- * Wear eye protection throughout and ensure that the lab is well ventilated.
- * Iron powder, Fe(s), (potential IRRITANT) – this can cause severe irritation in eyes because the iron oxidises rapidly in the saline environment
- * Iron powder is preferred to iron filings. If fine sulfur powder is mixed with iron filings, it is difficult to obtain a consistent mix, because the two solids can separate.
- * Sulfur, S(s) Roll sulfur or flowers of sulfur should be finely powdered using a pestle and mortar.
- * Sulfur dioxide, SO₂(g), (TOXIC) is formed if the sulfur catches fire. So, to prevent this, use cotton wool to plug the test tube.
- * This reaction can be carried out as a demonstration or class experiment in a well ventilated laboratory provided that the instructions provided here are strictly adhered to.
- *The reaction can be carried out in borosilicate glass test tubes as a demonstration or in smaller (ignition) tubes by students.
- *The reaction must not be carried out on tin lids in the open laboratory as is suggested in some sources. The sulfur may boil or burn releasing sulfur dioxide which is a TOXIC and CORROSIVE gas and may trigger an asthmatic attack.

RECOMMENDATION

CONCLUSION:

EXERCISE/ACTIVITY

Using a bar magnet, check and comment on the magnetic property of the product form.